$h(t) = e^{-2|t|} \quad x(t) = u(t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau = \int_{-\infty}^{\infty} x(t-\tau)\, h(\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} u(t-\tau)\, e^{-2|\tau|}\, d\tau = \int_{-\infty}^{t} e^{-2|\tau|}\, d\tau$$

$t<0$: $y(t) = \int_{-\infty}^{t} e^{2\tau} d\tau = \frac{1}{2} e^{2t} = \frac{e^{2t}}{2}$

$t>0$: $y(t) = \int_{-\infty}^{0} e^{2\tau} d\tau + \int_{0}^{t} e^{-2\tau} d\tau = \frac{1}{2} - \frac{1}{2} e^{-2t} + \frac{1}{2}$

$$= \frac{2 - e^{-2t}}{2}$$

$x(t) \rightarrow$ period $= 1$, $x(t) = |t| \quad -1/2 \le t \le 1/2$, $a_k = ?$

$x(t) = -t \quad t \in (-1/2, 0)$; $t \quad t \in (0, \frac{1}{2})$

$$\rightarrow a_k = \int_{-1/2}^{1/2} x(t)\, e^{-jk\omega_0 t} dt$$

$$= \int_{-1/2}^{0} (-t)\, e^{-jk\omega_0 t} dt + \int_{0}^{1/2} t\, e^{-jk\omega_0 t} dt$$

$$= \int_{0}^{1/2} t\, e^{jk\omega_0 t} dt + \int_{0}^{1/2} t\, e^{-jk\omega_0 t} dt$$

$$= \int_0^{1/2} t \cdot 2\cos(k\omega_0 t)\, dt$$

$$= \int_0^{1/2} \frac{2t}{k\omega_0}\, d(\sin k\omega_0 t)$$

$$= \frac{2t}{k\omega_0}\sin k\omega_0 t \Big|_0^{1/2} - \frac{2}{k\omega_0}\int_0^{1/2} \sin k\omega_0 t\, dt$$

$$= \frac{2}{(k\omega_0)^2}\cos(k\omega_0 t)\Big|_0^{1/2} = \frac{2}{k^2\omega^2}[\cos\pi k - 1]$$

$$= \frac{1}{2k^2\pi^2}[(-1)^k - 1]$$

$$a_0 = \int_{-1/2}^{1/2} x(t)\, dt = \frac{1}{2}$$

$y(t) = x_1(t-2) * x_2(-t+3)$

$x_1(t) = e^{-2t}u(t) \quad x_2(t) = e^{-3t}u(t) \quad y(s) = ?$

$$L\{y(t)\} = L\{x_1(t-2)\}\; L\{x_2(-t+3)\}$$

$$= L\{x_1(t)\}.e^{2t}.\; L\{x_2(t)\}e^{-3t}$$

$$= x_1(s).e^{2s}.\; x_2(-s)e^{-3s}$$

$$= x_1(s).x_2(-s)e^{-s}$$

$$X_1(S) = \frac{1}{S+2} \qquad Re\{S\} > -2$$

$$x_2(S) = \frac{1}{S+3} \qquad Re\{S\} > -3$$

$$\Rightarrow Y(S) = \frac{e^{-S}}{(S+2)(S+3)} \qquad Re\{S\} > -2$$

$x(t) \rightarrow$ 2 sided , poles at $\sigma = -3, -1$, zero at $\sigma = -2$; $X(S) = ?$ $x(t) = ?$, ROC = ?

ROC: $Re\{S\} \in [-3, -1]$

$$X(S) = \frac{S+2}{(S+1)(S+3)} = \frac{1}{2}\frac{1}{S+1} + \frac{1}{2}\frac{1}{S+3}$$

$$x(t) = \frac{1}{2}e^{-3t}u(t) - \frac{1}{2}e^{-t}u(-t)$$

$x(t) = e^{-2t}\sin(\omega_0 t)\,u(t)$ $\quad X(j\omega) = ?$

$$x(j\omega) = \int_{-\infty}^{\infty} x(t)\,e^{-j\omega t}\,dt$$

$$= \int_{-\infty}^{\infty} e^{-2t}\sin(\omega_0 t)u(t)\,e^{-j\omega t}\,dt$$

$$= \int_{0}^{\infty} \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}\,e^{-(2+j\omega)t}\,dt$$

$$= \frac{1}{2j} \int_0^\infty e^{-(2+j\omega-j\omega_0)t} - e^{-(2+j\omega+j\omega_0)t} \, dt$$

$$= \frac{1}{2j} \left[\frac{1}{2+j\omega-j\omega_0} - \frac{1}{2+j\omega+j\omega_0} \right]$$

$$= \frac{-j}{2} \frac{2(j\omega_0)}{(2+j\omega)^2 - (j\omega_0)^2}$$

$$X(j\omega) = \frac{\omega_0}{4-\omega^2+\omega_0^2+4j\omega}$$